

AP Question Paper 2016
Class 12 Physics, Paper-II

Time allowed : 3 hours

Maximum Marks : 60

SECTION-A

(10 × 2 = 20)

Note:i) Answer all the questions.

ii) Each question carries two marks.

iii) All are very short answer type questions.

1. What is Hypermetropia? How can it be corrected?

Ans: Hypermetropia (long-sightedness) are common eye conditions that occur when light does not focus on the retina of the eye. It can be corrected by using prescription glasses or contact lenses or in mild cases by laser eye surgery.

Treatments: Contact lens etc.

2. How do you convert a moving coil galvanometer into a voltmeter?

Ans: By connected with a high resistance, we can converted the moving coil galvanometer into the voltmeter. The sum of potential difference across galvanometer or potential difference across the high resistance is the potential difference across the given load resistance.

$$V = I_g(G+R)$$

3. Magnetic lines form continuous closed loops. why?

Ans: Magnetic field lines always form closed loop. because magnetic monopole does not exists(it always exists in pairs i.e, north and south together) therefor magnetic lines of force originating from north always end in south making a loop.

4. Define magnetisation of a sample . What is its SI unit?

Ans: Magnetization, also termed as magnetic polarization, is a vector quantity that gives the measure of the density of permanent or induced dipole moment in a given magnetic material. ... Magnetization of a given sample material M , can be defined as the net magnetic moment for that material per unit volume.

SI units of magnetisation is Ampere per metre i.e. Am^{-1} .

5. What is the phenomenon involved in the working of a transformer?

Ans. The phenomenon involved in the working of transformer are:

- Electromagnetic induction is the principle on which a transformer worked in the form of mutual induction.
- Generally the two coils, primary and secondary by connected together form the transformer.
- Mutual induction is the refers to the process by which a coil of wire induces a current into the neighbouring coil.
- Due to induction, voltage is induced in the secondary coil.

6. Give two uses of infrared Rays.

Ans: We use infrared radiation in many ways

- thermal imaging cameras
- remote controls

7. Write down de Broglie's relation and explain the terms there in.

Ans : $\lambda = h/mv$

where λ = the de Broglie wavelength (m)

h = Planck's constant ($6.63 \times 10^{-34} \text{ J.s}$)

p = momentum of a particle (kg. m/s)

m = mass of a particle (kg)

v = velocity of a particle (m/s)

8. The work function of cesium is 2.14 eV . Find the threshold frequency for cesium.

(Take $h = 6.6 \times 10^{-34}$ Js).

Ans - work function = 2.14 eV = $2.14 \times 1.6 \times 10^{-19}$ J

h = given

ν_0 = work function / h

$$\Rightarrow 2.14 \times 1.6 \times 10^{-19} / 6.6 \times 10^{-34}$$

$$\Rightarrow 5.16 \times 10^{14} \text{ Hz} \quad (5.16 \times 10^{14} \text{ Hz})$$

9. In which bias can a zener diode be used as voltage regulator?

Ans. A zener diode can be used as voltage regulator in the reverse bias. A regulator is provide the constant output to load which is connected in the parallel with it.

10. Define Modulation. Mention the basics methods of modulation.

Ans: Modulation is the process of converting data into radio waves by adding information to an electronic or optical carrier signal.

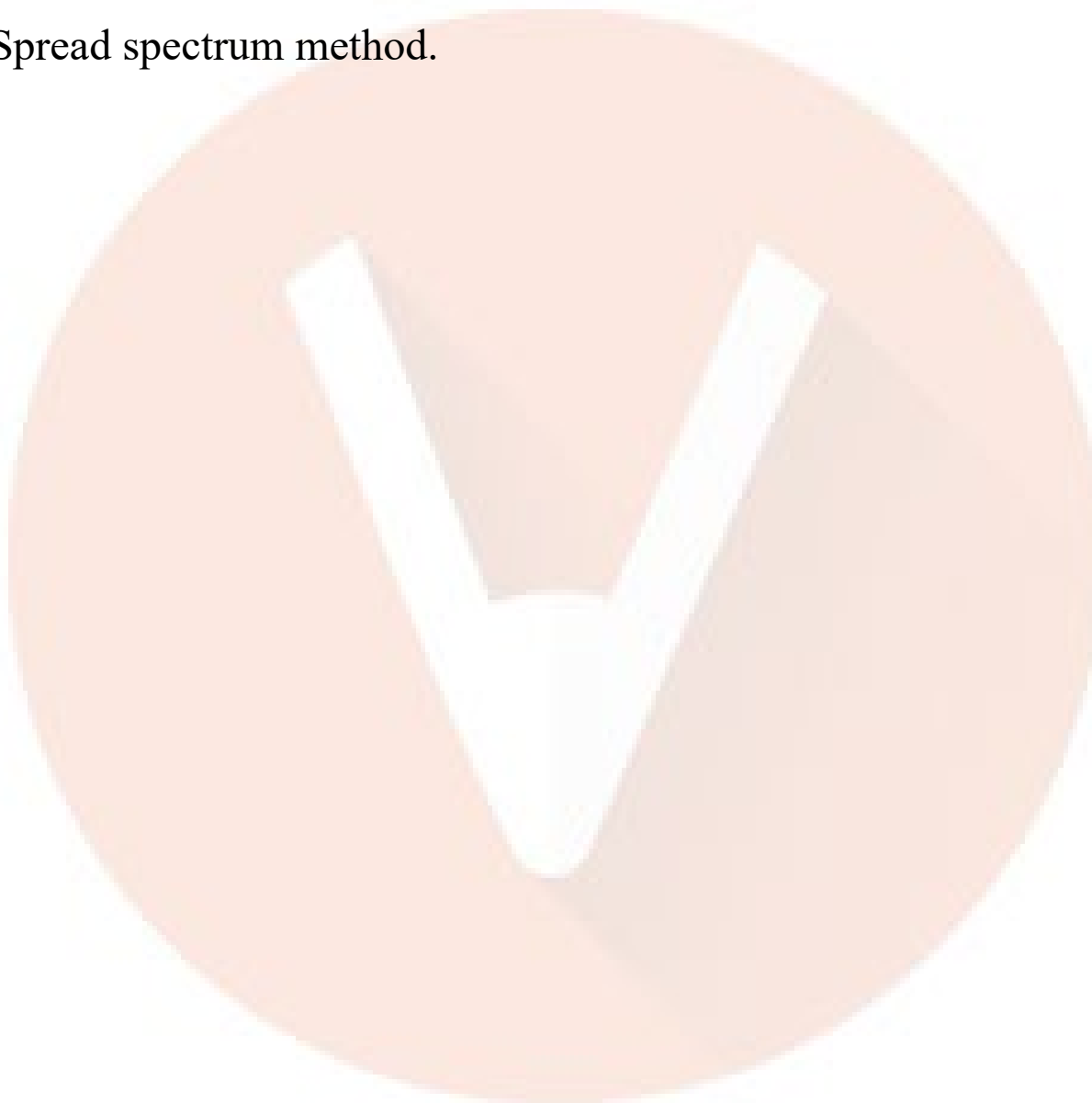
Modulation techniques are roughly divided into four types:

Analog modulation

Digital modulation

Pulse modulation

Spread spectrum method.



SECTION- B**(6 × 4 = 24)**

Note: i) Answer Any Six of the following questions.

ii) Each question carries four marks.

iii) All are short answer type questions

11. Explain the formation of mirage.

Ans. The formation of mirage occurs when the land is very hot and the air is cool. A hot land warms the layer of air which is just above the land. So, when the light moves through cold to hot air it is bent or refracted, while light travels in the straight line. When the light is bent, it is incident at an angle which is greater than the critical angle. At this time it suffers from the total internal reflection and is reflected upwards. As a result a mirage is formed.

12. Derive the expression for the intensity at a point where interference of light occurs. Arrive at the conditions for maximum and zero intensity.

Ans -Let Y_1 and Y_2 be displacement of two waves having same amplitude a and phase difference ϕ between them.

$$Y_1 = a \sin \omega t.$$

$$Y_2 = a \sin (\omega t + \phi)$$

Resultant displacement is $y = Y_1 + Y_2$

$$Y = a \sin \omega t + a \sin (\omega t + \phi) = a \sin \omega t (1 + \cos \phi) + \cos \omega t (a \sin \phi)$$

$$R \cos \theta = a (1 + \cos \phi)$$

$$Y = R \sin(\omega t + \theta)$$

Where,

R

R is resultant amplitude at p, i is intensity, squaring the equations we get,

$$I = R^2 = a^2 (1 + \cos \phi)^2 + a^2 (\sin \phi)^2$$

$$= 2a^2 (1 + \cos \phi) = 4a^2 \cos^2 \phi / 2$$

Maximum intensity:-

$$\cos^2 \phi / 2 = 1$$

$$\phi = 2n\pi \text{ where}$$

$$n = 0, 1, 2, 3,$$

Therefore,

$$I_{\max} = 4a^2$$

Minimum intensity:

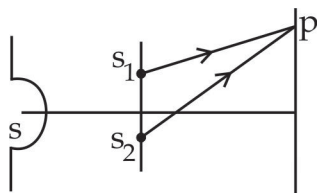
$$\cos^2 \phi / 2 = 0$$

$$\phi = (2n+1)\pi \text{ where}$$

$$n = 0, 1, 2, 3,$$

Therefore,

$$I_{\max} = 0$$



13. Define intensity of electric field at a point . Derive an expression for the intensity due to point charge.

Ans: Electric intensity is the strength of electric field at a point. Electric intensity at a point is defined as the force experienced per unit positive charge at a point placed in the electric field or It may also be also defined as the electrostatic force per unit.

Consider a point charge q called SOURCE CHARGE placed at a point 'O' in space. To find its intensity at a point 'p' at a distance 'r' from the point charge we place a test charge q .

The force experienced by the test charge q' will be:

$$F = Eq' \text{----(1)}$$

According to coulomb's law the electrostatic force between them is given by:

$$F = \frac{Kqq'}{r^2}$$

We know that

$$E = \frac{F}{q'}$$

Or

$$E = \frac{I}{q'} \times F$$

$$E = \frac{I}{q'} \times \frac{Kqq'}{r^2}$$

Putting the value of 'F' we get ,

$$E = \frac{Kq}{r^2}$$

But,

$$K = \frac{1}{4\pi\epsilon_0}$$

$$\therefore E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

This shows that the electric intensity due to a point charge is directly proportional to the magnitude of charge q and inversely proportional to the square of distance.

14. Derive an expression for the capacitance of a parallel plate capacitor.

Ans- The most common capacitor consists of two parallel plates. First, we derive the capacitance which depends on the area of the plates A and their separation d. According to

Gauss's law, the electric field between two plates is:

$$E = \frac{Q}{\epsilon_0 A} \Rightarrow Ed = V = \frac{Qd}{\epsilon_0 A}$$

Since the capacitance is defined by

$$V = \frac{Q}{C}$$

one can see that capacitance is:

$$C = \frac{\epsilon_0 A}{d}$$

If a dielectric material is inserted between the plates, the microscopic dipole moments of the material will shield the charges on the plates and alter the relation. Materials have a permeability ϵ which is sometimes given by the relative permeability k , $\epsilon = k\epsilon_0$. The capacitance is then given by:

$$C = \frac{\epsilon A}{d} = \frac{k\epsilon_0 A}{d}$$

15. Derive an expression for the magnetic dipole moment of a revolving electron.

Ans. Expression for the magnetic dipole moment of a revolving electrons:

Circulating electrons constitute a current,

$$I = e/T = e/2\pi R/v = ev/2\pi R$$

Magnetic moment associated with it,

$$\vec{u}l = iA$$

$$= ev / 2\pi R \cdot \pi R^2$$

$$= evR/2 \dots \dots \dots \text{equation 1}$$

Angular momentum associated with it, $l = mvR$

$$vr = l/m$$

Putting the value in equation 1, we get

$$\vec{u}l = el/2m \text{ or,}$$

$$\vec{u}/l = e/2m$$

16. Describe the ways in which Eddy current are used to advantage.

Ans: Eddy currents are loops of electrical current induced within conductors by a changing magnetic field in the conductor according to Faraday's law of induction. Eddy currents flow in closed loops within conductors in planes perpendicular to the magnetic field.

They are used to perform non-contact heating as in metal refineries, stove top containers. They can slow the movement of a moving conductor without contact. An example might be a spinning disc such as in a wheel of a vehicle. It could be used to perform heating for thermal stabilization in satellite or probes far from a warming source such as a star with long duration

operation without resorting to a mechanical means.

17. State the basic postulates of bohr's theory of atomic spectra.

Ans : Postulates of Bohr's Model of an Atom:-

1. atom has a nucleus where all protons and neutrons are present.
2. The size of the nucleus is very small and is present at the centre of the atom.
3. negatively charged electrons revolve around the nucleus.
4. the electrostatic force b/w nucleus and electron is equal to the centrifugal force on electron.
5. electrons can revolve only in those orbits in which angular momentum of electron is integral multiple of $\frac{h}{2\pi}$ i.e $mvr = \frac{nh}{2\pi}$; $n = +ve$ integral no., $h =$ planck's const.

18. Describe how a semiconductor diode is used as a half wave rectifier.

Ans. A semiconductor diode is used as the half wave rectifier:

At the time of positive half cycle a diode is set up under the forward bias condition and during this time it conducts current to load resistance. As in the input AC signal a voltage is also developed across the load. Just opposite to the positive half cycle, during the time of negative half cycle a diode is set up under the reverse bias condition and during this time it does not

conduct the current through the load resistance. Only AC input voltage appears across the load.

It is the Total result which is possible during the time of positive half cycle .



SECTION –C

(2 × 18 =16)

Note: i) Answer ANY TWO of the following questions.

ii) Each question carries Eight marks.

iii) All are long answer type questions

19. a) What is doppler effect ? Obtain an expression for the apparent frequency of sound heard when the source is in motion with respect to an observer at rest.

Ans : The Apparent change in the frequency of sound when the source, the observer and the medium are in a relative motion is called Doppler Effect.

Let S be source of sound moving with velocity v_s towards stationary listener O.

Distance travelled in time time period, $T = v_s T$

Therefore apparent wavelength, $\lambda' = \lambda - v_s T$

$$\lambda = \lambda' - v_s T$$

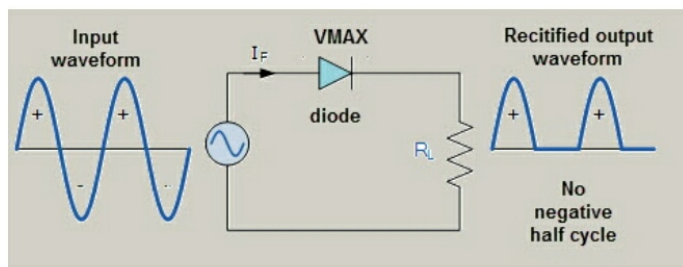
If apparent frequency heard by listener is v' .

$$v' = v / \lambda' = v v / v - v_s$$

Thus apparent frequency is greater than the actual frequency as source moves towards listener. Similarly if source moves away from listener.

$$v' = v/\lambda'$$

$$= vv/v + v_s$$



19. b) A rocket is moving at a speed of 200 ms^{-1} towards a stationary target. While moving, it emits a wave of frequency 1000 Hz . Calculate the frequency of the sound as detected by the target.

Ans : (1) The observer is at rest and the source is moving with a speed of 200 m s^{-1} . Since this is comparable with the velocity of sound, 330 m s^{-1} , we must use Eq. (15.50) and not the approximate Eq. (15.51). Since the source is approaching a stationary target, $v_o = 0$, and v_s must be replaced by $-v_s$. Thus, we have

$$v = v_0(1 - v_s/v)^{-1} \quad v = v_0(1 - v_s/v)^{-1}$$

$$v = 1000 \text{ Hz} \times [1 - 200 \text{ ms}^{-1} / 330 \text{ ms}^{-1}]^{-1} \quad v = 1000 \text{ Hz} \times [1 - 200 \text{ ms}^{-1} / 330 \text{ ms}^{-1}]^{-1}$$

$$\cong 2540 \text{ Hz} \cong 2540 \text{ Hz}$$

(2) The target is now the source (because it is the source of echo) and the rocket's detector is now the detector or observer (because it detects echo). Thus, $v_s = 0$ and v_o has a

positive value. The frequency of the sound emitted by the source (the target) is v , the frequency intercepted by the target and not v_0 . Therefore, the frequency as registered by the rocket is

$$v' = v(v + v_0/v)$$

$$= 2540 \text{ Hz} \times (200 \text{ ms}^{-1} + 330 \text{ ms}^{-1} / 330 \text{ ms}^{-1}) = 2540 \text{ Hz} \times (200 \text{ ms}^{-1} + 330 \text{ ms}^{-1} / 330 \text{ ms}^{-1})$$

$$\cong 4080 \text{ Hz}$$

20.a) State the working principle of potentiometer . Explain with help of circuit diagram how the emf of two primary cell are compared by using the potentiometer.

Ans : potentiometer works on the principle of galvanometer.

when we keep key (k_1) closed and (k_2) open ,let the null point found be I_{a1}

$$E_1 = K I_{a1} \dots\dots (1)$$

when we keep k_1 open and k_2 closed , let null point obtained by I_{a2} .

$$E_2 = K I_{a2} \dots\dots\dots (2)$$

$$(1)/(2)$$

$$\text{therefore } E_1/E_2 = I_1/I_2$$

where E_1 and E_2 are the two EMF of the given two cell.

21. Explain the principle and working of a nuclear reactor

with the help of a labelled diagram.

Ans: Nuclear reactors operate on the principle of nuclear fission, the process in which a heavy atomic nucleus splits into two smaller fragments. The nuclear fragments are in very excited states and emit neutrons, other subatomic particles, and photons.

Nuclear reactors are the heart of a nuclear power plant. They contain and control nuclear chain reactions that produce heat through a physical process called fission. That heat is used to make steam that spins a turbine to create electricity.